

Internship Report
Of
CODE-ALPHA
FRONT-END DEVELOPMENT INTERNSHIP

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: [25SCS1003004011]

Name of Student: ASHUTOSH SAHOO

Batch: 2025-29

2026-27

1. CANDIDATE'S DECLARATION

I, ASHUTOSH SAHOO, do hereby declare that the internship report titled “CodeAlpha Frontend Development Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: ASHUTOSH SAHOO

Roll

No.:

[25SCS1003004011]

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the CodeAlpha Internship Program. The internship provided me with practical exposure to frontend development and helped me apply classroom concepts through task-based projects using HTML, CSS and JavaScript.

I am thankful to CodeAlpha for providing the opportunity to work on practical frontend tasks and develop projects that improved my understanding of web interface design, responsiveness and JavaScript-based interaction. I am also thankful to IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continued support and encouragement.

I express my gratitude to my parents for their blessings and support throughout the internship and my academic journey.

Signature: _____

Student Name: ASHUTOSH SAHOO

Roll

No.:

[25SCS1003004011]

TABLE OF CONTENTS

Chapter No.	Chapter Name	Page
1.	Candidate's Declaration	i
2.	Acknowledgement	ii
3.	Internship Completion Certificate	-
4.	Project Description	1
4.1	Introduction	1
4.2	Organization Profile	2
4.3	Problem Statement	3
4.4	Project Objectives	3
4.5	Scope of the Project	4
4.6	Technologies and Tools Used	4
4.7	System Architecture	5
4.8	Methodology	6
4.9	Expected Outcomes	7
4.10	Certificates of Completion and Communication Proof	8
5.	Bibliography/References	9

3. INTERNSHIP COMPLETION CERTIFICATE

 CODE ALPHA YOUR DREAM OUR PASSION	CERTIFICATE OF COMPLETION	 Student ID: CA/DF1/237853
This Certificate Is Proudly Presented To <i>Ashutosh Sahoo</i> Was an active Participant at CodeAlpha Virtual Internship Program in Frontend Development Internship with dedication and hard work. We really appreciate your efforts taken and wish you all the best for the further. 10th August 2026 to 10th September 2026		
 CEO & Founder	11th September 2026 Date of Issue	 MINISTRY OF MICRO, SMALL & MEDIUM ENTERPRISES

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the frontend development internship completed under CodeAlpha. The internship was structured as a task-based learning experience in which practical web development concepts were applied through two completed projects: First-Calculator and Lumina-Image-gallery. The work focused on HTML, CSS and JavaScript, with emphasis on building functional interfaces, responsive layouts and basic interactive behaviour.

The internship provided a progression from a functionality-focused calculator to a visually focused image gallery with richer user interaction. Alongside development work, the projects were organized in public GitHub repositories and deployed using GitHub Pages, providing practical exposure to basic project management and static web hosting.

The two projects were therefore used as practical exercises to strengthen core frontend development skills while maintaining a simple and appropriate technology stack.

4.2 Organization Profile

CodeAlpha is identified in the provided internship context as a provider of a mini internship / frontend development task-based internship. The internship centered on practical beginner-level frontend tasks and the completion of web projects using HTML, CSS and JavaScript.

The available internship material does not provide a broader organizational history, management structure or detailed corporate profile. Accordingly, this report intentionally limits the organization profile to information that can be supported by the supplied internship records rather than introducing unsupported company details.

4.2 Organization Profile (continued)

Within the internship, the work environment was primarily project-oriented. The completed tasks emphasized the implementation of browser-based interfaces and the practical use of GitHub for repository management and GitHub Pages for deployment. This approach allowed the internship work to remain focused on fundamental frontend development while still producing publicly accessible outcomes.

4.3 Problem Statement

Students learning web development often understand individual HTML, CSS and JavaScript concepts separately but have limited experience combining them into complete, usable interfaces. The practical challenge addressed during this internship was to apply these technologies together to build functional and responsive web projects while gaining familiarity with basic development workflow and deployment.

For the internship, this challenge was approached through two small projects with increasing interface complexity. First-Calculator focused on implementing a functional interactive interface, while Lumina-Image-gallery extended the learning into responsive visual presentation, category-based filtering and image navigation.

4.4 Project Objectives

- Understand and apply basic HTML page structure for browser-based applications.
- Use CSS to create clean layouts, visual styling and responsive behaviour.
- Implement JavaScript-based interaction for functional user interfaces.
- Build an interactive calculator as a basic frontend application.
- Develop an image gallery with category filtering, navigation and enlarged image viewing.
- Practice DOM-based UI behaviour, hover effects and simple transitions.
- Organize projects in public GitHub repositories and deploy static websites using GitHub Pages.

4.3 Problem Statement / 4.4 Project Objectives (continued)

The objectives were approached incrementally so that each project introduced additional frontend concepts without requiring advanced frameworks or backend infrastructure. This kept the development process aligned with the beginner-level scope of the internship while still producing complete, shareable web applications.

4.5 Scope of the Project

The scope of the internship was limited to frontend development using HTML, CSS and JavaScript. The work covered the creation of static browser-based applications, responsive interface design and basic JavaScript interaction. The internship did not require a backend system, database, external application programming interface or advanced web framework.

The first project covered calculator interface construction, styling and JavaScript interaction. The second project covered responsive gallery layout, category-based filtering, navigation, enlarged image viewing, previous/next controls, hover effects and transitions. Both projects also included basic repository organization and deployment through GitHub Pages.

4.6 Technologies and Tools Used

Category	Technology / Tool	Use in Internship
Markup	HTML	Page structure and UI elements for both projects
Styling	CSS	Visual styling, layout and responsive behaviour
Programming	JavaScript	User interaction, calculations, filtering and navigation
Repository	GitHub	Public project repositories and source management
Deployment	GitHub Pages	Hosting the finished static websites

4.5 Scope of the Project (continued)

The project scope also included basic testing and refinement of the interfaces across different screen sizes. Since both applications were static frontend projects, their primary focus was the quality of the browser interface, interaction flow and responsiveness rather than server-side functionality.

4.6 Technologies and Tools Used (continued)

HTML, CSS and JavaScript formed the complete technical foundation of the two projects. GitHub was used for repository management and GitHub Pages was used to publish the completed static websites. Keeping the technology stack small made it possible to concentrate on core frontend concepts such as structure, styling, responsiveness and interaction.

The image content used in Lumina-Image-gallery was sourced from Unsplash, as recorded in the internship context. The project therefore combined locally implemented interface behaviour with externally sourced photography used for presentation purposes.

Project Summary

Project	Primary Focus	Key Features / Outcome
First-Calculator	Functional frontend interface	Responsive calculator layout, CSS styling and JavaScript interaction
Lumina-Image-gallery	Responsive visual interface	Cars, Flowers and Landscapes categories; filtering; navigation; enlarged viewing; hover effects and transitions

4.7 System Architecture

Because both projects are simple static frontend applications, their architecture is intentionally lightweight. The user interacts with a browser-based interface whose structure is provided by HTML, presentation is controlled by CSS, and interactive behaviour is implemented through JavaScript. The final files are maintained in GitHub repositories and published through GitHub Pages.



Figure 4.1 Front-end Architecture of First-Calculator

For First-Calculator, the flow begins with user input and passes through the HTML structure, CSS styling and JavaScript interaction before producing the browser-based calculator interface. The completed project is maintained in GitHub and published using GitHub Pages.



Figure 4.2 Front-end Architecture of Lumina-Image-gallery

For Lumina-Image-gallery, the flow begins with the gallery interface and category selection. JavaScript manages filtering and image controls, leading to enlarged image viewing and navigation. HTML and CSS provide the responsive presentation, while GitHub Pages publishes the finished static website.

4.8 Methodology

The internship followed a practical project-based methodology. Each project moved from understanding the task to interface construction, styling, JavaScript interaction, testing, repository preparation and deployment. The sequence below summarizes the development progression without assigning unsupported dates or weekly schedules.

Phase / Task	Description
1. Requirement understanding	Reviewed the project objective and identified the interface and interaction requirements.
2. Calculator interface	Created the HTML structure and visible controls for First-Calculator.
3. Styling and responsiveness	Applied CSS styling and responsive layout principles to the calculator.
4. JavaScript interaction	Implemented calculator behaviour and user interaction using JavaScript.
5. Testing and refinement	Checked the interface and refined layout or interaction behaviour as needed.
6. Repository and deployment	Prepared the public GitHub repository and deployed the calculator through GitHub Pages.
7. Gallery planning	Defined the category-based layout and visual structure for Lumina-Image-gallery.
8. Gallery implementation	Built the gallery interface, categories, filtering and navigation using HTML, CSS and JavaScript.
9. Image viewing interaction	Implemented enlarged image viewing and previous/next navigation.
10. Visual refinement	Applied hover effects, transitions and responsive behavior.
11. Final testing	Checked the gallery across screen sizes and refined the user interface.
12. Repository and deployment	Prepared the public GitHub repository and deployed the gallery through GitHub Pages.

4.8 Methodology (continued)

The methodology emphasized learning by implementation. Instead of treating HTML, CSS and JavaScript as isolated topics, the projects required their combined use within complete browser-based interfaces. The first project established the fundamentals of interaction and layout, while the second project introduced a broader set of UI behaviours such as filtering, navigation and image viewing.

A further part of the methodology was basic project delivery. After implementation and testing, both projects were maintained in public GitHub repositories and published using GitHub Pages. This provided practical exposure to the final steps involved in making a static web project accessible beyond the local development environment.

Development Progression

Stage	Project	Learning Progression
Project 1	First-Calculator	Focused on core functionality, HTML/CSS structure and JavaScript interaction.
Project 2	Lumina-Image-gallery	Extended the learning into responsive visual presentation and richer JavaScript-driven UI interaction.

4.9 Expected Outcomes

- Improved understanding of HTML page structure and frontend organization.
- Better practical use of CSS for styling, spacing, layouts and responsiveness.
- Improved understanding of JavaScript-based user interaction and DOM-driven behaviour.
- Experience in building small interactive web applications from start to deployment.
- Basic understanding of filtering, navigation, image viewing and UI transitions.
- Practical experience with GitHub repositories and GitHub Pages deployment.
- Greater confidence in continuing frontend development beyond beginner-level projects.

4.10 Certificates of Completion and Communication Proof

The official IILM report format requires the internship completion certificate and relevant communication proof to be attached as supporting evidence. The CodeAlpha Certificate of Completion supplied for this report is attached in Chapter 3. The certificate identifies Ashutosh Sahoo (Student ID: CA/DF1/237853) as an active participant in the CodeAlpha Virtual Internship Program in Frontend Development Internship and records the internship period as 10th August 2026 to 10th September 2026, with a date of issue of 11th September 2026.

5. BIBLIOGRAPHY/REFERENCES

1. CodeAlpha, “CodeAlpha Internship Context,” internship project documentation and project details, 2026.
 2. CodeAlpha, “Certificate of Completion,” Student ID: CA/DF1/237853, issued 11 September 2026.
 3. GitHub, “GitHub Documentation,” <https://docs.github.com/>
 4. GitHub, “GitHub Pages Documentation,” <https://docs.github.com/en/pages>
 5. MDN Web Docs, “HTML: HyperText Markup Language,” <https://developer.mozilla.org/en-US/docs/Web/HTML>
 6. MDN Web Docs, “CSS: Cascading Style Sheets,” <https://developer.mozilla.org/en-US/docs/Web/CSS>
 7. MDN Web Docs, “JavaScript,” <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
 8. Unsplash, image source platform used for photography in Lumina-Image-gallery, <https://unsplash.com/>
- First-Calculator - GitHub: <https://github.com/ashu2sh-hub/First-Calculator>
- First-Calculator - Live: <https://ashu2sh-hub.github.io/First-Calculator/>
- Lumina-Image-gallery - GitHub: <https://github.com/ashu2sh-hub/Lumina-Image-gallery>
- Lumina-Image-gallery - Live: <https://ashu2sh-hub.github.io/Lumina-Image-gallery/>